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Learning Sparsity Pattern: An Adaptive Joint Detection for AFDM Grant-Free System in LEO Satellites Communication

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***Abstract*—In this paper, an adaptive sparse pattern learning aided joint activity detection and channel estimation scheme is proposed for grant-free random access with affine frequency division multiplexing (AFDM) in LEO satellite communication systems. Specifically, the channel parameters and active index will be detected in the context of unknown sparsity. First, a sparse prior is preset to perform active detection and channel estimation. Subsequently, the prior is refined by a graph network architecture, where the delay-Doppler grid is modeled as the graph nodes delivering the message. Furthermore, a novel graph aggregation module based on self-attention mechanism is designed to capture the correlation between nodes to infer the unknown sparsity patterns inherent in dynamic terrestrial-satellite links (TSL). Simulation results demonstrate that, compared with conventional nearest neighbor sparsity pattern learning (NNSPL), the proposed scheme achieves a 2 dB improvement in normalized mean-square error (NMSE) and reduces pilot overhead by approximately 25%.**

***Index Terms*—Affine frequency division multiplexing (AFDM), grant-free random access (GF-RA), joint active detection and channel estimation (JADCE), graph neural network (GNN).**

I. Introduction

LOW earth orbit (LEO) satellites communication, enabling global coverage and ubiquitous connectivity, have been envisaged as a fundamental component of future Internet of Things (IoT) [1]. Over the next decade, large-scale constellations comprising thousands of LEO satellites will be deployed to serve billions of IoT devices [2]. However, owing to massive device connectivity and long terrestrial-satellite link (TSL) distances, conventional grant-based access mechanism gives rise to prohibitive transmission latency [3]. Accordingly, grant-free random access (GF-RA), which significantly reduces signaling overhead and access latency by streamlining the access process, constitutes as a promising alternative for LEO satellite IoT.

In GF-RA systems, each user transmits data directly to LEO satellites without prior scheduling. Nevertheless, the high Doppler shift caused by the mobility of LEO satellites poses severe interference for data transmission. To address this challenge, orthogonal time frequency space (OTFS) modulation is incorporated into GF-RA systems [4]. However, the high complexity of the 2D transformation in OTFS limits its practical deployment. In contrast, affine frequency division multiplexing (AFDM), based on 1D chirp modulation, is capable of achieving comparable performance with lower complexity [5]. Owing to these attractive features, AFDM emerges as a promising candidate for low-complexity and reliable GF-RA systems.

To unleash the full potential of AFDM-GF-RA, accurate joint activity detection and channel estimation (JADCE) is a critical prerequisite. By exploiting the sporadic sparsity of IoT device, JADCE has been widely formulated as a compressed sensing (CS) problem [6]–[12]. Specifically, representative algorithms including approximate message passing (AMP) [6], orthogonal matching pursui (OMP) [7], sparse Bayesian learning (SBL) [8], have attracted considerable attention in terrestrial networks owing to their efficient implementation. However, such approaches fail to exploit the block sparsity induced by the large delay and high Doppler characteristics of TSLs. As a remedy, advanced schemes have been developed to facilitate block sparse pattern through structured prior modeling or clustering methods, such as pattern coupled SBL [9], sparsity pattern learning (SPL) [10], [11], and modified variance state propagation [12]. However, since their convergence performance depends on predefined hyperparameters, the applicability of these schemes remains constrained, making it difficult to fully exploit the evolving structural sparsity in TSLs.

Benefiting from powerful feature extraction capabilities, deep learning-based schemes have been investigated for JADCE to alleviate the dependence on empirical models [13]–[15]. Representative works include data-driven [13] and model-driven methods [14], [15]. However, data-driven schemes suffer from limited interpretability, while model-driven approaches exhibit limited gains due to restricted parameterization. Moreover, owing to the absence of explicit modeling for complex sparsity patterns, these architectures are unable to exploit structural correlations, resulting in degraded performance in AFDM-GF-RA systems.

Against the above background, the contributions of this paper are summarized as follows:

1) We propose a graph neural network-based adaptive sparsity pattern learning (GNN-ASPL) scheme for AFDM-GF-RA system. Specifically, it extends the conventional sparse clustering methods into a modified message passing paradigm, and employs GNN to infer the unknown sparse pattern in an adaptive manner.

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2) An innovative graph aggregation module with self-attention is further designed to captures correlations among signal components to mitigate error propagation in sparse pattern estimation.
3) Moreover, the GNN model is iteratively integrated with a deep unfolding network to improve detection performance. Simulation results demonstrate that the proposed method significantly outperforms existing baselines in both accuracy and robustness.

Notations: $(\cdot)^{\mathrm{T}}$ and $(\cdot)^{\mathrm{H}}$ denote the transpose and Hermitian transpose, respectively. The notation $\mathbb{C}$ and $\mathbb{R}$ represent the complex and real number fields. $\mathcal{CN}(\mu,\sigma^2)$ denotes the complex Gaussian distribution with mean μ and variance σ^2, while $\sim$ denotes "is distributed as."

II. AFDM-GF-RA SYSTEM

We consider an AFDM-GF-RA system, where a LEO satellite equipped with M antennas serves K single-antenna devices. In each GF-RA round, only K_a devices are activated and the k-th device transmits signal $\mathbf{x}_k$ using AFDM modulation with N sub-carriers and sub-carrier spacing Δf.

The time-domain transmitted signal is generated through the inverse discrete affine Fourier transform (IDAFT) applied to $\mathbf{x}_k$, mathematically formulated as

$$s_k[n]=\frac{1}{\sqrt{N}}\sum_{m=0}^{N-1}x_k[m]\,e^{j2\pi\left(c_1n^2+c_2m^2+\frac{nm}{N}\right)}, \tag{1}$$

where $x_k[m]$ represents the m-th symbol of $\mathbf{x}_k$. c_1 and c_2 are AFDM modulation parameters. The signal in (1) can be represented in matrix form as

$$\mathbf{s}_k=\mathbf{A}^{\mathrm{H}}\mathbf{x}_k=\mathbf{\Lambda}_{c_1}^{\mathrm{H}}\mathbf{F}\mathbf{\Lambda}_{c_2}^{\mathrm{H}}\mathbf{x}_k, \tag{2}$$

where $\mathbf{A}^{\mathrm{H}}=\mathbf{\Lambda}_{c_1}^{\mathrm{H}}\mathbf{F}\mathbf{\Lambda}_{c_2}^{\mathrm{H}}$ represents the DAFT matrix. $\mathbf{F}\in\mathbb{C}^{N\times N}$ denotes the normalized DFT matrix, and $\mathbf{\Lambda}_c\triangleq\mathrm{diag}(e^{-j2\pi cn^2},n=0,1,\cdots,N-1)$ represents the affine transformation matrix.

After incorporating the chirp-periodic prefix (CPP), the transmitted signal propagates through the terrestrial satellite links (TSLs), which can be modeled by

$$\tilde{h}_{k,m}=\sum_{l=0}^{P-1}h_{k,m,l}\delta\left(\nu-\nu_{k,l}\Delta f\right)\delta\left(\tau-\tau_{k,l}T_s\right), \tag{3}$$

where $T_s=1/(N\Delta f)$ gives the sampling interval. $\tilde{h}_{k,m}$ represents the channel between the k-th device and m-th antenna. P denotes the number of multi-paths. $h_{k,m,l}$, $\nu_{k,l}$, and $\tau_{k,l}$ denote the channel gain, the normalized Doppler and time delay of the l-th path, respectively.

For the TSL channel, the above parameters are modeled according to 3GPP recommendations [16]. Specifically, the normalized Doppler $\nu_{k,l}=\alpha_{k,l}+\beta_{k,l}$, with $\alpha_{k,l}\in\{-\alpha_{\max},\ldots,\alpha_{\max}\}$ and $\beta_{k,l}\in(-\frac{1}{2},\frac{1}{2}]$ denoting the integer and fractional parts, respectively. In order to support massive access, a sufficiently large transmission bandwidth is adopted. Accordingly, the normalized delay is approximated as an integer, with $\tau_{k,l}\in\{0,1,\ldots,\tau_{\max}\}$.

After removing the CPP from the time-domain signal, the received signal at the m-th antenna can be written as

$$\tilde{\mathbf{y}}_m=\sum_{k=0}^{K-1}\sum_{l=0}^{P-1}\alpha_k h_{k,l,m}\mathbf{\Gamma}_{k,m,l}^{\mathrm{cpp}}\mathbf{\Delta}_{\nu_{k,l}}\mathbf{\Pi}_{l_{k,l}}\mathbf{s}_k+\tilde{\mathbf{n}}_m, \tag{4}$$

where $\alpha_k\in\{0,1\}$ denotes the activity state of the k-th device. $\mathbf{\Delta}_{\nu_{k,l}}=\mathrm{diag}\left(e^{-j2\pi\nu_{k,l}n/N},n=0,\cdots,N-1\right)$ represents the Doppler matrix. $\mathbf{\Pi}_{l_{k,l}}$ denotes the cyclic shift matrix, $\tilde{\mathbf{n}}_m$ represents the additive white Gaussian noise (AWGN) with variance σ^2.

Finally, the received signal can be obtained after the DAFT operation, as

$$\begin{aligned}\mathbf{y}_m&=\mathbf{A}\mathbf{r}_m,\\&=\sum_{k=0}^{K-1}\sum_{l=0}^{P-1}\alpha_k h_{k,m,l}\underbrace{\mathbf{A}\mathbf{\Gamma}_{k,m,l}^{\mathrm{cpp}}\mathbf{\Delta}_{\nu_{k,l}}\mathbf{\Pi}_{l_{k,l}}\mathbf{A}^H}_{\mathbf{\Phi}(\nu_{k,m,l},\tau_{k,m,l})}\mathbf{x}_k+\mathbf{A}\tilde{\mathbf{n}}_m,\end{aligned} \tag{5}$$

where $\mathbf{\Phi}\left(\nu_{k,m,l},\tau_{k,m,l}\right)$ characterizes joint delay-Doppler effects in TSLs.

III. PROPOSED JOINT DETECTION FOR AFDM-GF-RA

A. Pilot Arrangement and Dual Sparsity Model

For the k-th device, the transmitted symbol is arranged by

$$x_k[n]=\begin{cases}x_k^p[n], & n\in[Q,L)\ ,\\ 0, & n\in[0,Q)\cup[L,L+Q)\ ,\\ x_k^d[n], & \text{otherwise},\end{cases} \tag{6}$$

where $L=Q+L_p$, with $Q\triangleq(\tau_{\max}+1)(2(\alpha_{\max}+k_v)+1)$ representing the length of the zero guard interval adjacent to the pilot block and L_p denoting the pilot length.

According to the symbol arrangement, the channel response can be isolated from the surrounding data symbols at the receiver [5]. Consequently, the input-output relation associated with pilot can be expressed as

$$\mathbf{y}_m^p=\sum_{k=0}^{K-1}\sum_{l=0}^{P-1}\alpha_k h_{k,m,l}\mathbf{\Phi}^p\left(\tau_{k,m,l},\nu_{k,m,l}\right)\mathbf{x}_k^p+\mathbf{n}_m^p, \tag{7}$$

where $\mathbf{\Phi}^p\triangleq\mathbf{\Phi}[\mathcal{I}_r,\mathcal{I}_p]$, with the column index $\mathcal{I}_p=[Q,L-1]$ and the row index $\mathcal{I}_r=[\alpha_{\max}+k_v,\alpha_{\max}+k_v+L-1]$.

To facilitate estimation, the time delay and Doppler frequency deviation are uniformly divided into N_ι and N_κ grids, respectively. Let $N_g\triangleq(N_\iota+1)(N_\kappa+1)$ denote the total number of virtual delay-Doppler grids. Consequently, (7) is approximated as

$$\begin{aligned}\mathbf{y}_m^p&=\sum_{k=0}^{K-1}\sum_{l=0}^{P-1}\alpha_k\sum_{i=0}^{N_\iota}\sum_{j=0}^{N_\kappa}\iota_{k,i}\kappa_{k,j}\tilde{h}_{k,m}^{i,j}\mathbf{\Phi}^p\left(\tau_i,\nu_j\right)\mathbf{x}_k^p+\mathbf{n}_m^p\\&=\sum_{k=0}^{K-1}\tilde{\mathbf{\Psi}}_k\tilde{\mathbf{\Lambda}}_k\tilde{\mathbf{h}}_{k,m}+\mathbf{n}_m^p=\mathbf{\Psi}\mathbf{\Lambda}\mathbf{h}_m+\mathbf{n}_m^p,\end{aligned} \tag{8}$$

where $\iota_{k,i},\kappa_{k,j}\in\{0,1\}$ denotes the indicator of the virtual grid. $\tilde{h}_{k,m}^{i,j}$ and $\mathbf{\Phi}^p\left(\tau_i,\nu_j\right)$ denote the channel coefficient and the sub-channel matrix at the grid points, with $\tau_i=i\tau_{\max}/N_\iota$ and $\nu_j=(2j-N_\kappa)\,\nu_{\max}/N_\kappa$. $\mathbf{\Psi}=[\tilde{\mathbf{\Psi}}_0,\cdots,\tilde{\mathbf{\Psi}}_{K-1}]\in\mathbb{C}^{L\times KN_g}$ represents the expanded pilot matrix of all devices, with $\tilde{\mathbf{\Psi}}_k=[\mathbf{\Phi}^p\left(\tau_1,\nu_1\right)\mathbf{x}_k^p,\cdots,\mathbf{\Phi}^p\left(\tau_{N_\iota},\nu_{N_\kappa}\right)\mathbf{x}_k^p]$. $\mathbf{\Lambda}=\mathrm{diag}[\alpha_0\tilde{\mathbf{\Lambda}}_0,\cdots,\alpha_{K-1}\tilde{\mathbf{\Lambda}}_{K-1}]$ is the diagonal matrix, with $\tilde{\mathbf{\Lambda}}_k=\mathrm{diag}[\iota_{k,0}\kappa_{k,0},\cdots,\iota_{k,N_\iota}\kappa_{k,N_\kappa}]$. $\mathbf{h}_m=[\tilde{\mathbf{h}}_{0,m}^{\mathrm{T}},\cdots,\tilde{\mathbf{h}}_{K-1,m}^{\mathrm{T}}]^{\mathrm{T}}$ denotes the channel gains of all devices in different grids, with $\tilde{\mathbf{h}}_{k,m}=[\tilde{h}_{k,m}^{0,0},\cdots,\tilde{h}_{k,m}^{N_\iota,N_\kappa}]^{\mathrm{T}}$.

Combining the observed signals from multiple antennas, we have

$$\mathbf{Y}=\mathbf{\Psi}\mathbf{\Lambda}\mathbf{H}+\mathbf{N}=\mathbf{\Psi}\mathbf{Z}+\mathbf{N}, \tag{9}$$

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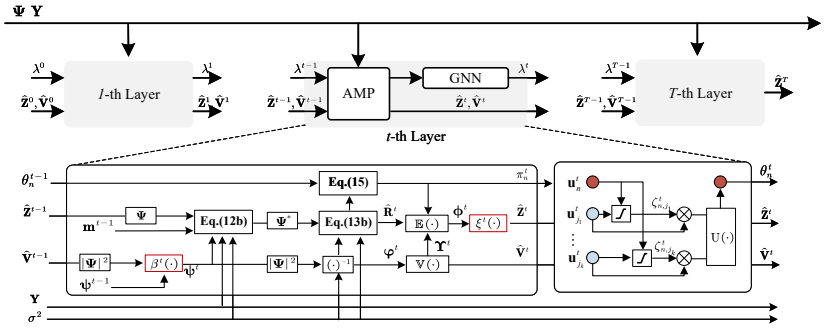


Fig. 1. The Architecture of GNN-ASPL

where $\mathbf{Z} = [(\mathbf{z}_0^{0,0})^{\mathrm{T}}, \cdots, (\mathbf{z}_k^{i,j})^{\mathrm{T}}, \cdots, (\mathbf{z}_K^{N_\epsilon, N_\kappa})^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{C}^{KN_g \times M}$ denotes the structured sparse channel model, with $\mathbf{z}_k^{i,j} = \alpha_k \iota_{k,i} \kappa_{k,j} \tilde{\mathbf{h}}_k^{i,j}$ and $\tilde{\mathbf{h}}_k^{i,j} = [\tilde{h}_{k,0}^{i,j}, \cdots, \tilde{h}_{k,M-1}^{i,j}]^{\mathrm{T}}$. Accordingly, $\mathbf{Z}$ exhibits a dual sparsity structure. Specifically, there are only K_a device blocks are nonzero due to α_k, while only P nonzero delay-Doppler grid are occupied within each active block.

According to the channel characteristics and the inherent dual sparse structure, $\mathbf{Z}$ is modeled by a hierarchical spike-and-slab prior, as

$$\begin{aligned} p(\mathbf{Z}|\boldsymbol{\lambda}) &= \prod_{k=0}^{K-1} \prod_{i,j=0}^{N_g-1} \left((1-\lambda_k^{i,j}) \delta(\mathbf{z}_k^{i,j}) + \lambda_k^{i,j} \mathcal{CN}(\mathbf{z}_k^{i,j}; \boldsymbol{\mu}_k^{i,j}, \boldsymbol{\Sigma}_k^{i,j}) \right) \\ p(\boldsymbol{\lambda}; \boldsymbol{\theta}) &= \prod_{k=0}^{K-1} \prod_{i,j=0}^{N_g-1} \left((1-\theta_k(\mathcal{G}_k^{i,j})) \delta(\lambda_k^{i,j}) + \theta_k(\mathcal{G}_k^{i,j}) \delta(1-\lambda_k^{i,j}) \right), \end{aligned} \tag{10}$$

where $\delta(\cdot)$ denotes the Dirichlet function. $\boldsymbol{\mu}_k^{i,j}$ and $\boldsymbol{\Sigma}_k^{i,j} = \mathrm{diag}([\gamma_{k,0}^{i,j}, \cdots, \gamma_{k,M-1}^{i,j}])$ denote the mean and variance, respectively. $\boldsymbol{\lambda}$ denotes the latent support vector of $\mathbf{Z}$, with $\theta_k(\mathcal{G}_k^{i,j})$ representing the structured sparse probability of $\lambda_k^{i,j}$. Specifically, $\theta_k(\cdot)$ captures device-level inter-group sparsity, while $\mathcal{G}_k^{i,j}$ controls intra-group delay-Doppler sparsity.

Within the Bayesian framework, the maximum a posteriori (MAP) estimate can be formulated as

$$\begin{aligned} \hat{\mathbf{Z}}, \hat{\boldsymbol{\theta}} &= \arg\max_{\mathbf{Z}, \boldsymbol{\theta}} p(\mathbf{Z}|\mathbf{Y}; \boldsymbol{\theta}) \\ &= \arg\max_{\mathbf{Z}, \boldsymbol{\theta}} p(\mathbf{Y}|\mathbf{Z})\, p(\mathbf{Z}|\boldsymbol{\lambda})\, p(\boldsymbol{\lambda}; \boldsymbol{\theta}). \end{aligned} \tag{11}$$

However, direct estimation of these two parameters poses considerable challenges, since the the structured sparse probability $\theta_k(\mathcal{G}_k^{i,j})$ is generally difficult to specify at the receiver. To address this issue, an intelligent estimation framework is designed and will be detailed in the next subsection.

B. Proposed GNN-ASPL Algorithm

In this subsection, a GNN-ASPL algorithm is proposed to exploit the dual sparse structure of $\mathbf{Z}$. Compared to conventional sparsity pattern learning (SPL) schemes [11], which rely on predefined sparse associations, the proposed detector constructs the latent support variables $\theta_k(\mathcal{G}_k^{i,j})$ as graph nodes and employs GNN-based message aggregation to learn their structured dependencies. In this way, the coupling between block sparsity induced by device activity and intra-block sparsity in the delay-Doppler domain can be adaptively captured. The architecture of GNN-ASPL is shown in Fig. 1, by unfolding the AMP algorithm and embedding the GNN model, GNN-ASPL enables efficient inference while adaptively learning the coupled sparse structure. Specifically, detailed procedure is presented as follows.

Initialization. For notational simplicity, (k, i, j) is indexed by n in the following derivations, and the sparse probability θ_n^0 is initialized as 0.5 to compute the posterior under a fixed sparse pattern.

Step 1: Update the variance $\psi_{k,m}^t$ and mean $m_{k,m}^t$. To initialize the subsequent sparse structure learning, the posterior statistics of $\mathbf{Z}$ are first obtained through iterative inference. At the t-th iteration, the variance $\psi_{k,m}^t$ and mean $m_{k,m}^t$ are updated as

$$\psi_{k,m}^t = \beta^t \sum_{n=0}^{KN_g-1} |\Psi_{k,n}|^2 V_{n,m}^{t-1} + \left(1-\beta^t\right) \psi_{k,m}^t \tag{12a}$$

$$m_{k,m}^t = \sum_{n=0}^{KN_g-1} \Psi_{k,n} \hat{z}_{n,m}^{t-1} + \frac{\psi_{k,m}^t \left(y_{k,m} - m_{k,m}^t\right)}{\sigma^2 + \psi_{k,m}^{t-1}} \tag{12b}$$

where β^t denotes the trainable damping factor of t-th layer.

Step 2: Update the variance $\varphi_{n,m}^t$ and observation $r_{n,m}^t$. Subsequently, the equivalent noise variance $\varphi_{n,m}^t$ and observation $r_{n,m}^t$ are obtained as

$$\varphi_{n,m}^t = \left(\sum_{k=0}^{L-1} \frac{|\Psi_{k,n}|^2}{\sigma^2 + \psi_{k,m}^t} \right)^{-1} \tag{13a}$$

$$r_{n,m}^t = \hat{z}_{n,m}^{t-1} + \zeta_{n,m}^t \sum_{k=0}^{L-1} \frac{\Psi_{k,n}^* \left(y_{k,m} - m_{k,m}^t\right)}{\sigma^2 + \psi_{k,m}^t}. \tag{13b}$$

Step 3: Compute the sparse estimate π_n^t. Based on the equivalent observation, the posterior distribution of $\mathbf{z}_n$ can be approximated as

$$p\left(\mathbf{z}_n|\hat{\mathbf{r}}_n^t\right) = \left(1-\pi_n^t\right) \delta\left(\mathbf{z}_n\right) + \pi_n^t \mathcal{CN}\left(\mathbf{z}_n; \boldsymbol{\phi}_n^t, \boldsymbol{\Upsilon}_n^t\right), \tag{14}$$

where $\hat{\mathbf{r}}_n^t$ denotes the n-th row of $\hat{\mathbf{R}}^t$, and π_n^t represents the posterior sparse probability, given by

$$\pi_n^t = \frac{\theta_n^{t-1} \eta_n^t}{\left(1-\theta_n^{t-1}\right) + \theta_n^{t-1} \eta_n^t}, \tag{15}$$

with

$$\begin{aligned} \eta_n^t = \exp\Bigg(& \underbrace{- \sum_{m=0}^{M-1} \frac{\left|\hat{r}_{n,m}^t - \mu_{n,m}\right|^2}{\varphi_{n,m}^t + \gamma_{n,m}}}_{u_1} + \underbrace{\sum_{m=0}^{M-1} \frac{\left|\hat{r}_{n,m}^t\right|^2}{\varphi_{n,m}^t}}_{u_2} \\ & \underbrace{- \sum_{m=0}^{M-1} \log\left(\frac{\varphi_{n,m}^t + \gamma_{n,m}}{\varphi_{n,m}^t} \right)}_{u_3} \Bigg), \end{aligned} \tag{16}$$

where η_n^t denotes the likelihood ratio between the nonzero

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and zero components, while $\{u_1, u_2, u_3\}$ denotes the dominant terms of η_n^t. $\boldsymbol{\phi}_n^t$ and $\boldsymbol{\Upsilon}_n^t$ denote the mean and variance of the posterior distribution in (14), which can be obtained by

$$\Upsilon_{n,m}^t = \frac{\gamma_{n,m}\varphi_{n,m}^t}{\gamma_{n,m} + \varphi_{n,m}^t} \tag{17a}$$

$$\phi_{n,m}^t = \Upsilon_{n,m}^t \left(\frac{1}{\gamma_{n,m}} \mu_{n,m} + \frac{1}{\varphi_{n,m}^t} \hat{r}_{n,m}^t \right) \tag{17b}$$

Step 4: Recalculate the variance $V_{n,m}^t$ and the mean $z_{n,m}^t$. Based on the approximate posterior distribution in (14), an MMSE estimator is employed, and the posterior mean and variance are calculated as

$$z_{n,m}^t = \xi^t \pi_n^t \phi_{n,m}^t + \left(1 - \xi^t\right) z_{n,m}^{t-1}, \tag{18a}$$

$$V_{n,m}^t = \pi_n^t \left(1 - \pi_n^t\right) |\phi_{n,m}^t|^2 + \pi_n^t \Upsilon_{n,m}^t. \tag{18b}$$

To ensure proper convergence of GNN-ASPL, trainable damping factors are added to (12a) and (18a) for tuning the AMP module to the GNN module.

Step 5: Attention-based Message Aggregation. To update sparsity hyperparameter $\hat{\boldsymbol{\theta}}^t$, the EM algorithm is employed with the approximate distribution obtained in (14), and the sparsity can be updated as

$$\theta_n^t = \pi_n^t. \tag{19}$$

However, the update in (19) only exploits the row sparsity in the delay-Doppler domain, while ignoring the block sparsity shared by each device. In fact, the sparsity within each device is jointly constrained by its activity state and the intra-block sparsity pattern. Accordingly, the sparsity of each node is related to those of its associated neighbors in the same block.

To exploit this correlation, (19) is extends into a general message passing paradigm, which can be formulated as

$$\theta_n^t = \mathrm{U}^t \left(\theta_n^{t-1}, \sum\nolimits_{j \in \mathbb{I}_n} \varpi_{j \to n}^t \left(\pi_n^t, \pi_j^t, e_{n,j} \right) \pi_j^t \right), \tag{20}$$

where $\mathbb{I}_n$ denotes the set of node indices potentially associated with n. $\varpi_{j \to n}(\cdot)$ denotes the association function of node j and node n, conditioned on node and edge features. $\mathrm{U}(\cdot)$ is the update function, responsible for fusing the prior representation and the aggregated messages.

It is worth noting that the update of node n is determined by both θ_n^{t-1} and the aggregated sparse evidence from its associated nodes. When $\varpi_{j \to n}^t(\cdot)$ assigns comparable weights to nodes within the same device block, the aggregated message reflects the common block activity, thereby capturing block sparsity. In contrast, when $\varpi_{j \to n}^t(\cdot)$ concentrates on a few nodes with stronger sparse evidence, the update is dominated by the significant delay-Doppler components, thereby characterizing intra-block sparsity.

Benefiting from the ability to assign feature-dependent weights to different neighboring nodes, the self-attention mechanism is adopted to adaptively learn $\varpi_{j \to n}^t(\cdot)$ for joint block-level aggregation and intra-block sparse selection.

Specifically, the sparse estimate π_n^t is treated as the prior information for node n. To characterize the sparsity of the current node, the three dominant terms $\{u_1, u_2, u_3\}$ in (16) are incorporated. Accordingly, the node property $\mathbf{u}_n^t$ can be obtained by

$$\mathbf{u}_n^t = \left[u_1, u_2, u_3, \pi_n^t \right]. \tag{21}$$

A self attention mechanism is subsequently employed to compute the correlation $e_{n,j}$ between nodes n and j, which can be represented by

$$e_{n,j} = \mathrm{LeakRelu}\left(\left[\mathbf{W}\mathbf{u}_n^t || \mathbf{W}\mathbf{u}_j^t \right] \mathbf{a} \right), \tag{22}$$

where $\mathbf{W} \in \mathbb{R}^{h_1 \times 4}$ is a shared trainable parameter for feature enhancement. $\mathbf{a} \in \mathbb{R}^{2h_1 \times 1}$ is a single-layer feedforward network, which projects the concatenated high-dimensional features onto a real value.

To measure the degree of association, the correlation coefficient of adjacent nodes can be normalized as

$$\zeta_{n,j} = \frac{\exp\left(e_{n,j}\right)}{\sum_{j \in \mathbb{I}_n} \exp\left(e_{n,j}\right)}. \tag{23}$$

By fusing the correlation degree and node characteristics, the message $\mathbf{m}_{j \to n}^t$ transmitted from node j to node n can be obtained by

$$\mathbf{m}_{j \to n}^t = \zeta_{n,j} \mathbf{u}_j^t. \tag{24}$$

In the following step, the n-th node aggregates all incoming messages and updates its feature using a gated recurrent unit (GRU) network.

$$\mathbf{g}_n^t = \mathrm{GRU}(\mathbf{g}_n^{t-1}, \sum_{j \in \mathbb{I}_n(n)} \mathbf{m}_{j \to n}^t), \tag{25}$$

where $\mathrm{GRU}(\cdot)$ denotes the GRU network. $\mathbf{g}_n^t \in \mathbf{R}^{h_2 \times 1}$ represents the hidden feature of the n-th node, which is forwarded to the next GRU unit for iteration.

Step 6: Sparse Update. Finally, the sparsity hyperparameter is updated through a linear projection followed by a Sigmoid function, i.e.,

$$\lambda_n^t = \mathrm{Sigmoid}\left(\mathbf{W}_2 \mathbf{g}_n^t + \mathbf{b} \right). \tag{26}$$

After the T-layer iterative recovery, the activity state of the k-th device is detected according to the normalized average magnitude of the corresponding recovered block, which is obtained by

$$\hat{\alpha}_k = \begin{cases} 1, & \dfrac{\frac{1}{N_g M} \sum\limits_{n=(k-1)N_g+1}^{kN_g} \|\hat{\mathbf{z}}_n\|^2}{\max\limits_{1 \le j \le K} \left(\frac{1}{N_g M} \sum\limits_{n=(j-1)N_g+1}^{jN_g} \|\hat{\mathbf{z}}_n\|^2 \right)} \ge \eta, \\ 0, & \text{otherwise}, \end{cases} \tag{27}$$

where $N_g = (N_\iota + 1)(N_\kappa + 1)$ denotes the block length associated with each device, while η is the decision threshold.

Overall, the proposed GNN-ASPL framework alternates between AMP-based channel estimation and GNN-based adaptive sparsity pattern learning across T independent layers. This iterative design significantly enhances detection performance under unknown sparsity conditions.

C. Offline Training of GNN-ASPL

To facilitate the joint optimization of AMP and GNN modules, an improved layer-wise training strategy is adopted. Specifically, the training process is divided into T stages, corresponding to the T layers of the network. In the t-th stage, the GNN module from layer $t-1$ is first trained to refine the sparsity prior, with parameters of other layers kept fixed. Subsequently, the t-th AMP module is optimized based on the updated prior. To avoid overfitting caused by partial

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training, all parameters up to layer t are jointly optimized in the following step.

In order to obtain desirable training results, the above procedure adopts the normalized mean squared error (NMSE) as the loss function, which is expressed as

$$\mathcal{L}(\mathbf{\Omega}) = \frac{1}{|\Pi|}\sum_{i=1}^{|\Pi|}\frac{\|\mathbf{Z}_i - \hat{\mathbf{Z}}_i^t\|_{\mathrm{F}}^2}{\|\mathbf{Z}_i\|_{\mathrm{F}}^2}, \tag{28}$$

where $\mathbf{\Omega}$ denotes the set of trainable parameters, and Π represents the mini-batch training set with cardinality $|\Pi|$.

D. Complexity Analysis

In this subsection, the computational complexity is evaluated in terms of the real-valued floating-point operations (FLOPs). Specifically, given matrix $\mathbf{A_c} \in \mathbb{C}^{m\times n}$, $\mathbf{B_c} \in \mathbb{C}^{n\times p}$, $\mathbf{A_r} \in \mathbb{R}^{m\times n}$, $\mathbf{B_r} \in \mathbb{R}^{n\times p}$, the operations of $\mathbf{A_cB_c}$, $\mathbf{A_rB_r}$ require $8mnp - 2mp$, $2mnp - mp$ flops, respectively. Accordingly, the complexity for the t-th layer of proposed GNN-ASPL can be expressed as

$$\begin{aligned}\text{FLOPs} = &\underbrace{10\left(KN_g+1\right)LM}_{Eq.(12)} + \underbrace{KN_g\left(13L+3\right)M}_{Eq.(13)} \\ &+\underbrace{KN_g\left(17M+7\right)}_{Eq.(15)} + \underbrace{11KN_gM}_{Eq.(17)} + \underbrace{18KN_gM}_{Eq.(18)} \\ &+\underbrace{KN_g\left[J^2+11h_1J+3J+4h_2^2+7h_2\right]}_{Eq.(22)-(26)},\end{aligned} \tag{29}$$

where J denotes the cardinality of the index set $\mathbb{I}$.

It is worth noting that the GNN-related complexity term is independent of the number of receive antennas and the pilot length. Instead, it is only determined by the graph structure and the neural-network dimensions. In particular, owing to the block-sparse property of the considered system, J is only confined within a single device block. Hence, its relative computational proportion decreases as the antenna dimension or pilot length increases, making it attractive for large-scale deployment.

IV. Simulation Results

A. Simulation Setup

This section considers a typical LEO-enabled GF-RA system, where a satellite serves $K = 100$ potential devices within its coverage area. Each device is activated with a probability of $p_k = 0.1$ and transmits data using the AFDM waveform with a carrier frequency of 4 GHz and a subcarrier spacing of 15 kHz. The non-terrestrial channel model recommended in [16] is adopted, with the number of paths $P = 3$. Due to coarse pre-compensation, the residual delay and Doppler shifts for each device are assumed within a small range, where the maximum normalization delay $\tau_{\max}$ is set to 2 and the maximum residual Doppler is set to 30 kHz. The delays and Doppler are assumed to be aligned with the grid points, with $(N_l, N_k) = (2, 4)$.

In the offline training stage, GNN-ASPL is implemented via Pytorch platform on Dell T640 with a NVIDIA RTX 3090 GPU. To achieve satisfactory performance, over 10,000 pairs of training data $(\mathbf{Y}, \mathbf{\Psi}, \mathbf{Z}, \sigma^2)$ are randomly generated at different SNRs and a ten-layer GNN-ASPL is trained for each SNR. The Adam optimizer is adopted, with an initial learning rate 5×10^{-3} and a batchsize of 128.

B. Performance Comparsion

To demonstrate the effectiveness of the proposed GNN-ASPL, the benchmark detectors using for comparison include vector AMP with soft thresholding shrinkage (VAMP-ST) [6], block orthogonal matching pursuit (BOMP) [7], generalized AMP based SBL (GAMP-SBL) [8], GAMP with EM (EM-GAMP) [10], and nearest neighbor SPL (NNSPL) [11], as well as oracle least squares (OLS), which constitutes a computable performance bound in [4]. The NMSE and active device error rate (ADER) in different SNR are adopted as the metrics for channel estimation and active detection, respectively, where $\text{ADER} = \sum_k^K |\hat{\alpha}_k - \alpha|/K$ and $\text{SNR} = ||\mathbf{SZ}||_F^2/(ML\sigma^2)$.

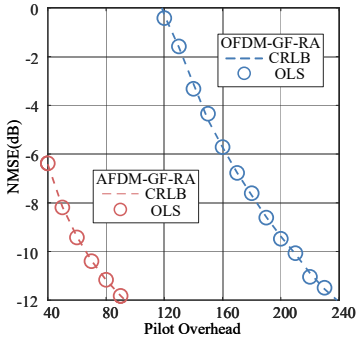


Fig. 2. The Theoretical performance for AFDM-GF-RA and OFDM-GF-RA systems with $(K, M, N_\iota, N_\kappa) = (100, 4, 2, 4)$ and SNR = 8 dB.

Fig. 2 compares the theoretical performance of the AFDM-GF-RA and OFDM-GF-RA systems, where the Cramer-Rao lower bound (CRLB) is derived under the assumption of known sparse support. It can be observed that the OLS closely coincide with the corresponding CRLBs, indicating that the OLS is consistent with the theoretical lower bound. Moreover, the results indicate that the AFDM-GF-RA requires significantly fewer pilots while achieving same NMSE, which demonstrates the superiority of AFDM in high mobility.

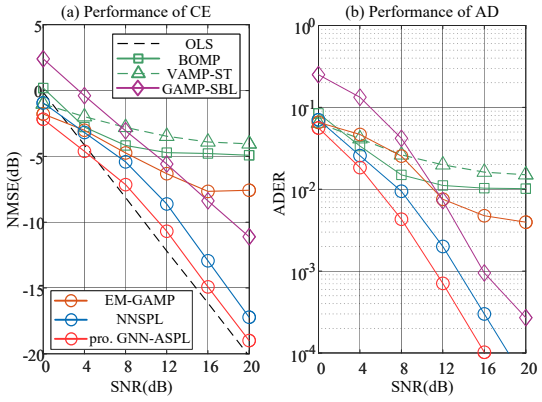


Fig. 3. The performance comparison of conventional detectors and proposed GNN-ASPL for AFDM-GF-RA system with $(K, M, N_\iota, N_\kappa) = (100, 4, 2, 4)$ and pilot overhead $L_p = 50$.

Fig. 3 depicts the NMSE and ADER achieved by the various competing schemes, where $(K, M, N_\iota, N_\kappa) = (100, 4, 2, 4)$ and the pilot overhead is $L_p = 50$. It can be observed from Fig. 3(a) that the proposed GNN-ASPL outperforms the benchmark schemes across the entire SNR range. Specifically, both SOMP

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and VAMP-ST suffer from an error floor, while GNN-ASPL is capable of achieving about 2 dB performance gain compared to NNSPL in terms of NMSE of -15 dB. This improvement is attributed to its capability to adaptively capture evolving sparsity patterns, as illustrated in Fig. 3(b), leading to effective suppression of error propagation during detection.

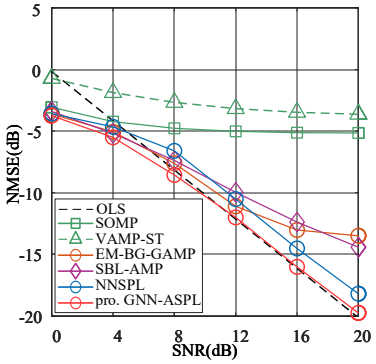


Fig. 4. The performance comparison of conventional detectors and proposed GNN-ASPL for AFDM-GF-RA system with $(K, M, N_\iota, N_\kappa) = (100, 32, 2, 4)$ and pilot overhead $L_p = 50$.

Fig. 4 depicts the NMSE performance of different schemes under the large-scale antenna configuration with $(K, M, N_\iota, N_\kappa) = (100, 32, 2, 4)$ and $L_p = 50$. It can be observed that the proposed GNN-ASPL achieves the best performance over the entire SNR range and closely approaches the OLS in the high-SNR region. By contrast, SOMP and VAMP-ST suffer from pronounced error floors, while EM-BG-GAMP and SBL-AMP exhibit limited improvement in the high-SNR region. Compared with NNSPL, GNN-ASPL provides about 1.8 dB NMSE improvement at SNR $= 20$ dB, which indicates its effectiveness in exploiting the spatial observations provided by massive antennas.

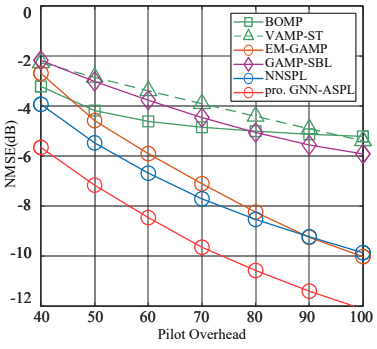


Fig. 5. The NMSE performance comparison with different pilot overhead with $(K, M, N_\iota, N_\kappa) = (100, 4, 2, 4)$ and SNR = 8 dB.

To provide further analysis, Fig. 5 compares the NMSE performance versus the number of pilots under SNR = 8 dB. As shown in Fig. 5, the proposed GNN-ASPL can significantly reduce pilot overhead. In particular, to achieve an NMSE of –10 dB, the proposed scheme requires only 75 pilots, while NNSPL needs 100 pilots, corresponding to a 25% reduction. This highlights the significant advantage of the proposed algorithm in resource-constrained scenarios.

V. Conclusion

In this paper, an AFDM-GF-RA system was investigated for LEO satellite IoT to mitigate excessive signaling overhead and access latency. To enable reliable uplink access, the JADCE problem was first formulated as a structured compressed sensing task. To address this problem, a GNN-ASPL algorithm was proposed. Specifically, a self-attention-based graph aggregation module was developed to refine sparsity patterns, which was further embedded into a deep unfolding framework to enhance sparse recovery. Simulation results verified the significant performance advantages of the proposed approach and demonstrated its strong generalization capabilities under dynamic TSL conditions.

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